

Radiation Safety (Diagnostic and Interventional)

Policy Number	POL-SAF-070	Owner / Department	Radiation Safety / Radiology
Effective Date	2023-09-01	Version	3.1
Last Revised	2026-03-12	Next Review	2028-03-12
Approved By	Radiation Safety Committee	Applies To	All personnel who operate, assist with, or work near ionizing-radiation sources, including radiology, interventional radiology, cardiac cath lab, nuclear medicine, and operating-room staff

1. Purpose and Scope

This policy establishes the facility radiation-protection program for all sources of ionizing radiation at Riverside Regional Medical Center, consistent with the **as-low-as-reasonably-achievable (ALARA)** principle. It applies to diagnostic x-ray, fluoroscopy, interventional procedures, and byproduct material used under the facility radioactive-materials license.

The program is administered by the **Radiation Safety Officer (RSO)** under the oversight of the Radiation Safety Committee and conforms to U.S. Nuclear Regulatory Commission standards for protection against radiation (10 CFR Part 20) and for the medical use of byproduct material (10 CFR Part 35), as adopted by the State radiation-control program.

2. Occupational Dose Limits

Occupational exposures shall be kept ALARA and shall not exceed the annual limits below (10 CFR 20.1201, 20.1207, 20.1208). Facility investigation levels are set at a fraction of the regulatory limit to prompt review before a limit is approached.

Exposure category	Annual limit	Facility action level
Total effective dose equivalent (whole body)	5 rem (50 mSv)	1.25 rem / quarter
Lens of the eye	15 rem (150 mSv)	3.75 rem / quarter
Skin / any extremity	50 rem (500 mSv)	12.5 rem / quarter
Minor (under 18)	10% of adult limits	Any measurable reading reviewed
Declared pregnant worker (gestation)	0.5 rem (5 mSv) to embryo/fetus	0.05 rem / month

A worker may **declare a pregnancy in writing** to the RSO to invoke the embryo/fetus limit; the declaration may be withdrawn at any time in writing.

3. Dosimetry and ALARA Controls

- Personnel likely to receive >10% of an annual limit are issued dosimeters; collar badges are worn outside the apron at the collar, with a second under-apron badge for declared pregnant workers.
- Dosimeters are exchanged and read monthly; results are reviewed by the RSO against facility action levels.
- Apply time, distance, and shielding: minimize beam-on time, maximize distance from the source, and use lead aprons, thyroid shields, and ceiling-suspended shields.
- Fluoroscopy operators are credentialed and monitor cumulative air-kerma and fluoroscopy time; prolonged procedures trigger a substantial-radiation-dose review.
- Radioactive materials are secured, labeled, surveyed, and disposed of per the license and 10 CFR 35.

4. Overexposure and Patient Dose Events

A suspected overexposure, lost or contaminated source, or unintended high patient dose must be reported immediately to the RSO, who determines regulatory reportability (e.g., a 10 CFR 35.3045 medical event) and State radiation-control notification obligations.

IMPORTANT — If a dosimeter reading or clinical finding suggests an overexposure or a lost/unsecured radioactive source, stop the activity, secure the area, and call the Radiation Safety Officer immediately (24/7 pager via operator). Do not resume use until the RSO clears the area.

5. Severity & Reporting

Classify and notify per the facility-wide severity scale in **POL-RM-013**. A radioactive-source exposure causing serious injury or a reportable medical event with major patient harm is a **SEV-1**; a significant protocol deviation or action-level exceedance without serious harm is a **SEV-2**; a minor variance or a Near Miss is **SEV-3 / Near Miss**.

Severity	Notify	Deadline
SEV-1 (catastrophic / sentinel)	RSO, Risk Management, administrator on call	Within 1 hour
SEV-2 (major)	RSO and department manager	Within 4 hours
SEV-3 / Near Miss (minor)	RSO via electronic event report	Within 24 hours

The RSO reports regulatory events to the State radiation-control program and NRC as required; staff-injury aspects are cross-referenced to the bloodborne/exposure policy **POL-EH-001** where applicable.

6. References

- 10 CFR Part 20 — Standards for Protection Against Radiation (NRC).
- 10 CFR Part 35 — Medical Use of Byproduct Material (NRC), including 35.3045 medical-event reporting.
- State radiation-control program regulations and the facility radioactive-materials license.
- POL-RM-013 — Incident Reporting and Event Severity Classification.
- POL-EH-001 — Bloodborne Pathogen Exposure and Needlestick.